

3 Theory of Quotients—First Principles of Theory of Numbers

Exercises III

1. Express in its simplest form—

$$\frac{x^2}{x-y} + \frac{y^2}{y-x}.$$

$$\begin{aligned} \frac{x^2}{x-y} + \frac{y^2}{y-x} &= \frac{x^2(y-x)}{(x-y)(y-x)} + \frac{y^2(x-y)}{(x-y)(y-x)} && \left(\frac{a}{b} + \frac{p}{q} = \frac{qa}{qb} + \frac{pb}{qb}\right) \\ &= \frac{x^2(y-x) + y^2(x-y)}{(x-y)(y-x)} && \left(\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}\right) \\ &= \frac{x^2(y-x) + y^2(x-y)}{xy - x^2 - y^2 + xy} && \text{(expand denominator)} \\ &= \frac{x^2(y-x) + y^2(x-y)}{x(y-x) + y(x-y)} && \text{(factor denominator)} \\ &= \frac{x^2(y-x) - y^2(y-x)}{x(y-x) - y(y-x)} && (x-y = -(y-x)) \\ &= \frac{(x^2 - y^2)(y-x)}{(x-y)(y-x)} && \text{(extract common factor } y-x) \\ &= \frac{(x^2 - y^2)}{(x-y)} && \left(\frac{ac}{bc} = \frac{a}{b}\right) \\ &= \frac{(x-y)(x+y)}{(x-y)} && (a^2 - b^2 = (a-b)(a+b)) \\ &= x+y && \left(\frac{ac}{bc} = \frac{a}{b}\right) \end{aligned}$$

2. Express in its simplest form—

$$\frac{a}{a-b} + \frac{b}{b-a}.$$

$$\begin{aligned} \frac{a}{a-b} + \frac{b}{b-a} &= \frac{a(b-a) + b(a-b)}{(a-b)(b-a)} && (\frac{a}{b} + \frac{p}{q} = \frac{aq}{bq} + \frac{bp}{bq}) \\ &= \frac{a(b-a) + b(a-b)}{ab - a^2 - b^2 + ab} && (\text{expand denominator}) \\ &= \frac{a(b-a) + b(a-b)}{a(b-a) + b(a-b)} && (\text{factor denominator}) \\ &= 1 && (\frac{a}{a} = 1) \end{aligned}$$

3. Simplify—

$$\frac{P+Q}{P-Q} - \frac{P-Q}{P+Q}, \quad \text{where } P = x+y, \quad Q = x-y.$$

$$\begin{aligned} \frac{P+Q}{P-Q} - \frac{P-Q}{P+Q} &= \frac{x+y+x-y}{x+y-x+y} - \frac{x+y-x+y}{x+y+x-y} && (\text{replace variables}) \\ &= \frac{2x}{2y} - \frac{2y}{2x} && (\text{simplify}) \\ &= \frac{x}{y} - \frac{y}{x} && (\frac{ab}{bc} = \frac{a}{c}) \\ &= \frac{x^2 - y^2}{xy} && (\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}) \end{aligned}$$

4. Simplify—

$$\frac{1 - \frac{x(1-y)}{x+y}}{1 + \frac{1-y}{x+y}}.$$

$$\begin{aligned}
1 - \frac{x(1-y)}{x+y} \div 1 + \frac{1-y}{x+y} &= \frac{x+y}{x+y} - \frac{x(1-y)}{x+y} \div \frac{x+y}{x+y} + \frac{1-y}{x+y} \\
&= \frac{x+y-x+xy}{x+y} \div \frac{x+y+1-y}{x+y} \\
&= \frac{xy+y}{x+y} \div \frac{x+1}{x+y} \\
&= \frac{xy+y}{x+y} \times \frac{x+y}{x+1} \\
&= \frac{xy+y}{x+1} \\
&= \frac{y(x+1)}{x+1} \\
&= y
\end{aligned}$$

5. Simplify—

$$\frac{\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc}}{\frac{a^2 - (b-c)^2}{a}}$$

$$\begin{aligned}
\frac{\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc}}{\frac{a^2 - (b-c)^2}{a}} &= \left(\frac{1}{ab} - \frac{1}{ac} - \frac{1}{bc} \right) \div \frac{a^2 - (b-c)^2}{a} && \left(\frac{a}{b} = a \div b \right) \\
&= \left(\frac{c}{abc} - \frac{b}{abc} - \frac{a}{abc} \right) \div \frac{a^2 - (b-c)^2}{a} && \text{(common denominator } abc) \\
&= \frac{c-b-a}{abc} \div \frac{a^2 - (b-c)^2}{a} && \left(\frac{a}{b} - \frac{c}{b} = \frac{a-c}{b} \right) \\
&= \frac{c-b-a}{abc} \div \frac{(a-(b-c))(a+(b-c))}{a} && (a^2 - b^2 = (a-b)(a+b)) \\
&= \frac{c-b-a}{abc} \div \frac{(a-b+c)(a+b-c)}{a} && \text{(remove brackets)} \\
&= \frac{c-b-a}{abc} \times \frac{a}{(a-b+c)(a+b-c)} && \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p} \right) \\
&= \frac{-(a+b-c)}{abc} \times \frac{a}{(a-b+c)(a+b-c)} && (c-b-a = -(a+b-c)) \\
&= \frac{-a(a+b-c)}{abc(a-b+c)(a+b-c)} && \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq} \right) \\
&= -\frac{1}{bc(a-b+c)} && \left(\frac{ac}{bc} = \frac{a}{b} \right)
\end{aligned}$$

6. Simplify—

$$\left(a - \frac{b^2}{a+b}\right) \times \left(a + \frac{b^2}{a-b}\right).$$

$$\begin{aligned} & \left(a - \frac{b^2}{a+b}\right) \times \left(a + \frac{b^2}{a-b}\right) \\ &= \left(\frac{a(a+b)}{a+b} - \frac{b^2}{a+b}\right) \times \left(\frac{a(a-b)}{a-b} + \frac{b^2}{a-b}\right) \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \\ &= \left(\frac{a^2+ab-b^2}{a+b}\right) \times \left(\frac{a^2-ab+b^2}{a-b}\right) \quad \left(\frac{a}{c} - \frac{b}{c} = \frac{a-b}{c}\right) \\ &= \frac{(a^2+ab-b^2)(a^2-ab+b^2)}{(a+b)(a-b)} \quad \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq}\right) \\ &= \frac{a^4 - a^2b^2 + 2ab^3 - b^4}{(a+b)(a-b)} \quad (\text{expand numerator}) \\ &= \frac{a^4 - a^2b^2 + 2ab^3 - b^4}{a^2 - b^2} \quad (\text{distribute denominator}) \end{aligned}$$

7. Simplify—

$$\frac{1}{x+y} + \frac{1}{x-y} - \frac{2x}{x^2+y^2}.$$

$$\begin{aligned} & \frac{1}{x+y} + \frac{1}{x-y} - \frac{2x}{x^2+y^2} \\ &= \frac{x-y+x+y}{(x+y)(x-y)} - \frac{2x}{x^2+y^2} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq}{bq} - \frac{pb}{bq}\right) \\ &= \frac{2x}{x^2-y^2} - \frac{2x}{x^2+y^2} \quad (\text{collect like terms; } (a+b)(a-b) = a^2 - b^2) \\ &= \frac{2x(x^2+y^2) - 2x(x^2-y^2)}{(x^2-y^2)(x^2+y^2)} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq-pb}{bq}\right) \\ &= \frac{2x^3 + 2xy^2 - 2x^3 + 2xy^2}{(x^2-y^2)(x^2+y^2)} \quad (\text{expand numerator}) \\ &= \frac{4xy^2}{x^4 + x^2y^2 - y^2x^2 - y^4} \quad (\text{collect like terms; expand denominator}) \\ &= \frac{4xy^2}{x^4 - y^4} \quad (\text{collect like terms}) \end{aligned}$$

8. Simplify—

$$\left(\frac{x}{y} + \frac{y}{x}\right) \left(\frac{a}{b} + \frac{b}{a}\right) - \left(\frac{x}{y} - \frac{y}{x}\right) \left(\frac{a}{b} - \frac{b}{a}\right).$$

$$\begin{aligned}
& \left(\frac{x}{y} + \frac{y}{x} \right) \left(\frac{a}{b} + \frac{b}{a} \right) - \left(\frac{x}{y} - \frac{y}{x} \right) \left(\frac{a}{b} - \frac{b}{a} \right) \\
&= \frac{x^2 + y^2}{xy} \times \frac{a^2 + b^2}{ab} - \frac{x^2 - y^2}{xy} \times \frac{a^2 - b^2}{ab} & \left(\frac{a}{b} \pm \frac{p}{q} = \frac{aq \pm bp}{bq} \right) \\
&= \frac{(x^2 + y^2)(a^2 + b^2)}{abxy} - \frac{(x^2 - y^2)(a^2 - b^2)}{abxy} & \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq} \right) \\
&= \frac{(x^2 + y^2)(a^2 + b^2) - (x^2 - y^2)(a^2 - b^2)}{abxy} & \left(\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b} \right) \\
&= \frac{a^2x^2 + b^2x^2 + a^2y^2 + b^2y^2 - (a^2x^2 - b^2x^2 - a^2y^2 + b^2y^2)}{abxy} & \text{(expand numerator)} \\
&= \frac{a^2x^2 + b^2x^2 + a^2y^2 + b^2y^2 - a^2x^2 + b^2x^2 + a^2y^2 - b^2y^2}{abxy} & \text{(remove brackets)} \\
&= \frac{2b^2x^2 + 2a^2y^2}{abxy} & \text{(collect like terms)} \\
&= \frac{2(b^2x^2 + a^2y^2)}{abxy} & \text{(extract common factor 2)}
\end{aligned}$$

9. Simplify—

$$\left(\frac{a}{b} - \frac{b}{a} \right) \div \left(\frac{a^2}{b^2} - \frac{b^2}{a^2} \right).$$

$$\begin{aligned}
& \left(\frac{a}{b} - \frac{b}{a} \right) \div \left(\frac{a^2}{b^2} - \frac{b^2}{a^2} \right) \\
&= \frac{a^2 - b^2}{ab} \div \frac{a^4 - b^4}{a^2b^2} & \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= \frac{a^2 - b^2}{ab} \times \frac{a^2b^2}{a^4 - b^4} & \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p} \right) \\
&= \frac{a^2 - b^2}{ab} \times \frac{a^2b^2}{(a^2 - b^2)(a^2 + b^2)} & (a^2 - b^2 = (a - b)(a + b)) \\
&= \frac{a^2b^2(a^2 - b^2)}{ab(a^2 - b^2)(a^2 + b^2)} & \left(\frac{a}{b} \times \frac{p}{q} = \frac{ap}{bq} \right) \\
&= \frac{ab}{a^2 + b^2} & \left(\frac{ac}{bc} = \frac{a}{b} \right)
\end{aligned}$$

10. Simplify—

$$\frac{a(a - b) - b(a + b)}{\frac{a}{a + b} - \frac{b}{a - b}}.$$

$$\begin{aligned}
& \frac{a(a-b) - b(a+b)}{\frac{a}{a+b} - \frac{b}{a-b}} \\
&= (a(a-b) - b(a+b)) \div \left(\frac{a}{a+b} - \frac{b}{a-b} \right) \quad \left(\frac{a}{b} = a \div b \right) \\
&= (a(a-b) - b(a+b)) \div \frac{a(a-b) - b(a+b)}{(a+b)(a-b)} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= (a(a-b) - b(a+b)) \times \frac{(a+b)(a-b)}{a(a-b) - b(a+b)} \quad \left(a \div \frac{p}{q} = a \times \frac{q}{p} \right) \\
&= (a+b)(a-b) \quad \left(\frac{ac}{bc} = \frac{a}{b} \right) \\
&= a^2 - b^2 \quad ((a+b)(a-b) = a^2 - b^2)
\end{aligned}$$

11. Simplify—

$$\frac{1-x}{1+x+x^2} - \frac{1+x}{1-x+x^2}.$$

$$\begin{aligned}
& \frac{1-x}{1+x+x^2} - \frac{1+x}{1-x+x^2} \\
&= \frac{(1-x)(1-x+x^2) - (1+x)(1+x+x^2)}{(1+x+x^2)(1-x+x^2)} \quad \left(\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq} \right) \\
&= \frac{1-2x+2x^2-x^3 - (1+2x+2x^2+x^3)}{(1+x+x^2)(1-x+x^2)} \quad (\text{expand numerator}) \\
&= \frac{-4x-2x^3}{(1+x^2)^2 - x^2} \quad ((p+q)(p-q) = p^2 - q^2) \\
&= \frac{-4x-2x^3}{1+x^2+x^4} \quad (\text{expand denominator})
\end{aligned}$$

12. Simplify—

$$\frac{1 + \frac{x}{1+x}}{x + \frac{1}{1+x}} \div \frac{(x+1)^2 - x^2}{x^2 + x + 1}.$$

$$\begin{aligned}
& \frac{1 + \frac{x}{1+x}}{x + \frac{1}{1+x}} \div \frac{(x+1)^2 - x^2}{x^2 + x + 1} \\
&= \frac{\frac{1+2x}{1+x}}{\frac{x^2+x+1}{1+x}} \div \frac{2x+1}{x^2+x+1} \quad (\text{common denominator } 1+x; \text{ expand}) \\
&= \frac{1+2x}{1+x} \times \frac{1+x}{x^2+x+1} \div \frac{2x+1}{x^2+x+1} \quad \left(\frac{a}{b} \div \frac{p}{q} = \frac{a}{b} \times \frac{q}{p}\right) \\
&= \frac{1+2x}{x^2+x+1} \times \frac{x^2+x+1}{2x+1} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right) \\
&= 1 \quad \left(\frac{a}{a} = 1\right)
\end{aligned}$$

13. Simplify—

$$\frac{a^2 + b^2}{(a+b)^2} + \frac{\frac{2}{ab} \left(\frac{1}{a} + \frac{1}{b}\right)}{\left(\frac{1}{a} + \frac{1}{b}\right)^3}.$$

$$\begin{aligned}
& \frac{a^2 + b^2}{(a+b)^2} + \frac{\frac{2}{ab} \left(\frac{1}{a} + \frac{1}{b}\right)}{\left(\frac{1}{a} + \frac{1}{b}\right)^3} \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{1}{\left(\frac{1}{a} + \frac{1}{b}\right)^2} \quad \left(\frac{ac}{bc} = \frac{a}{b}, \text{ cancelling } \frac{1}{a} + \frac{1}{b}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{1}{\left(\frac{a+b}{ab}\right)^2} \quad \left(\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2}{ab} \times \frac{a^2 b^2}{(a+b)^2} \quad \left(\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right) \\
&= \frac{a^2 + b^2}{(a+b)^2} + \frac{2ab}{(a+b)^2} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right) \\
&= \frac{a^2 + 2ab + b^2}{(a+b)^2} \quad \left(\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}\right) \\
&= \frac{(a+b)^2}{(a+b)^2} \quad (a^2 + 2ab + b^2 = (a+b)^2) \\
&= 1 \quad \left(\frac{a}{a} = 1\right)
\end{aligned}$$

14. Show that—

$$\frac{x^4}{a^2b^2} + \frac{(x^2 - a^2)^2}{a^2(a^2 - b^2)} - \frac{(x^2 - b^2)^2}{b^2(a^2 - b^2)}$$

is independent of x .

Proof. Taking $a^2b^2(a^2 - b^2)$ as a common denominator,

$$\frac{x^4}{a^2b^2} + \frac{(x^2 - a^2)^2}{a^2(a^2 - b^2)} - \frac{(x^2 - b^2)^2}{b^2(a^2 - b^2)} = \frac{N}{a^2b^2(a^2 - b^2)},$$

where

$$\begin{aligned} N &= x^4(a^2 - b^2) + b^2(x^2 - a^2)^2 - a^2(x^2 - b^2)^2 \\ &= x^4(a^2 - b^2) + b^2(x^4 - 2a^2x^2 + a^4) \\ &\quad - a^2(x^4 - 2b^2x^2 + b^4) && ((p - q)^2 = p^2 - 2pq + q^2) \\ &= a^2x^4 - b^2x^4 + b^2x^4 - 2a^2b^2x^2 + a^4b^2 \\ &\quad - a^2x^4 + 2a^2b^2x^2 - a^2b^4 && (\text{remove brackets}) \\ &= a^4b^2 - a^2b^4 && (\text{collect like terms}) \\ &= a^2b^2(a^2 - b^2) && (\text{extract common factor } a^2b^2) \end{aligned}$$

Then the expression is

$$\frac{a^2b^2(a^2 - b^2)}{a^2b^2(a^2 - b^2)} = 1.$$

From this we see that all terms of x cancel, so its value is always 1, thus the expression is independent of x . $\square$

15. Simplify—

$$\frac{a}{b - \frac{c}{d - \frac{e}{f}}}.$$

$$\begin{aligned}
& \frac{a}{b - \frac{c}{d - \frac{e}{f}}} \\
&= a \div \left[b - \left(c \div \left(d - \frac{e}{f} \right) \right) \right] & (\frac{a}{b} = a \div b) \\
&= a \div \left[b - \left(c \div \frac{df - e}{f} \right) \right] & (a - \frac{p}{q} = \frac{aq - p}{q}) \\
&= a \div \left[b - \frac{cf}{df - e} \right] & (a \div \frac{p}{q} = \frac{aq}{p}) \\
&= a \div \left[\frac{b(df - e) - cf}{df - e} \right] & (a - \frac{p}{q} = \frac{aq - p}{q}) \\
&= \frac{a(df - e)}{b(df - e) - cf} & (a \div \frac{p}{q} = \frac{aq}{p}) \\
&= \frac{adf - ae}{bdf - be - cf} & (\text{expand})
\end{aligned}$$

16. Simplify—

$$\frac{1}{a - 2b - \frac{1}{a - 2b - \frac{1}{a - 2b}}}$$

Writing $t = a - 2b$,

$$\begin{aligned}
& \frac{1}{a - 2b - \frac{1}{a - 2b - \frac{1}{a - 2b}}} \\
&= \frac{1}{t - \frac{1}{t - \frac{1}{t}}} && \text{(replace variables)} \\
&= \frac{1}{t - \frac{1}{\frac{t^2 - 1}{t}}} && (\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq}) \\
&= \frac{1}{t - \frac{t}{t^2 - 1}} && (a \div \frac{p}{q} = \frac{aq}{p}) \\
&= \frac{1}{\frac{t(t^2 - 1) - t}{t^2 - 1}} && (\frac{a}{b} - \frac{p}{q} = \frac{aq - pb}{bq}) \\
&= \frac{t^2 - 1}{t^3 - 2t} && \text{(expand; } a \div \frac{p}{q} = \frac{aq}{p} \text{)} \\
&= \frac{(a - 2b)^2 - 1}{(a - 2b)^3 - 2(a - 2b)} && \text{(restore variables)}
\end{aligned}$$

17. Simplify—

$$\frac{a + b}{a + b + \frac{1}{a - b + \frac{1}{a + b}}}.$$

$$\begin{aligned}
& \frac{a+b}{a+b+\frac{1}{a-b+\frac{1}{a+b}}} \\
&= \frac{a+b}{a+b+\frac{1}{(a-b)(a+b)+1}} \quad \left(\frac{a}{b} + \frac{p}{q} = \frac{aq+pb}{bq}\right) \\
&= \frac{a+b}{a+b+\frac{a+b}{a^2-b^2+1}} \quad ((a+b)(a-b) = a^2 - b^2; a \div \frac{p}{q} = \frac{aq}{p}) \\
&= \frac{a+b}{\frac{(a+b)(a^2-b^2+1) + (a+b)}{a^2-b^2+1}} \quad \left(\frac{a}{b} + \frac{p}{q} = \frac{aq+pb}{bq}\right) \\
&= \frac{a+b}{\frac{(a+b)(a^2-b^2+2)}{a^2-b^2+1}} \quad (\text{extract common factor } a+b) \\
&= \frac{(a+b)(a^2-b^2+1)}{(a+b)(a^2-b^2+2)} \quad \left(a \div \frac{p}{q} = \frac{aq}{p}\right) \\
&= \frac{a^2-b^2+1}{a^2-b^2+2} \quad \left(\frac{ac}{bc} = \frac{a}{b}\right)
\end{aligned}$$

Exercises IV

1. If the two fractions A/B , a/b be equal, and the latter be at its lowest terms, prove that $A = \mu a$, $B = \mu b$, where μ is an integer.

Proof. Suppose A , B , a , b integers with $A/B = a/b$, and a/b at its lowest terms; that is, a and b have no common measure save unity, so that a is prime to b (§6). Multiplying across by bB ,

$$Ab = aB;$$

hence a divides Ab , and being prime to b it must divide A (§10, Cor. 1). Write $A = \mu a$, μ an integer. Substituting, $\mu ab = aB$, and cancelling a (which is not zero), $B = \mu b$ —the same μ serving in both. $\square$

2. Prove that the sum or difference of two odd numbers is always even; the sum or difference of an odd and an even number always odd; the product of any number of odd numbers always odd; the quotient of one odd number by another always odd, if it be integral.

Proof. Let a, b, c be integers where a is even and b and c are odd. Then $a = 2m$, $b = 2n + 1$, and $c = 2l + 1$ for some integers m, n, l (Cor., §12). Next, we prove the propositions individually.

1. *Sum or difference of two odd numbers is always even.* Take $b + c$, the sum of two odd numbers, this can be written $b + c = (2n + 1) + (2l + 1) = 2n + 1 + 2l + 1 = 2n + 2l + 2 = 2(n + l + 1)$. Thus, the sum is always even. Now take the difference of two odd numbers, in either order: $b - c = (2n + 1) - (2l + 1) = 2n + 1 - 2l - 1 = 2n - 2l = 2(n - l)$, and $c - b = (2l + 1) - (2n + 1) = 2l + 1 - 2n - 1 = 2l - 2n = 2(l - n)$. Thus the difference is always even.
2. *Sum or difference of an odd and an even number is always odd.* Take $a + b$, the sum of an even number and an odd number, this can be written $a + b = 2m + (2n + 1) = 2(m + n) + 1$. Thus, the sum is always odd. Now take the difference, in either order: $a - b = 2m - (2n + 1) = 2(m - n - 1) + 1$, and $b - a = (2n + 1) - 2m = 2(n - m) + 1$. Thus the difference is always odd.
3. *The product of any number of odd numbers is always odd.* First take two of them, $bc = (2n + 1)(2l + 1) = 4nl + 2n + 2l + 1 = 2(2nl + n + l) + 1$, which is odd. Now suppose the product of k odd numbers to be odd, say $2p + 1$, and let $2q + 1$ be a further odd number. Then $(2p + 1)(2q + 1) = 2(2pq + p + q) + 1$, which is odd again. Hence the product of $k + 1$ odd numbers is odd whenever the product of k of them is; and since the product of two is odd, the product of any number of odd numbers is odd.
4. *The quotient of one odd number by another is always odd, if it be integral.* Let b/c be integral, say $b = cq$, where q is an integer. Were q even, $q = 2k$ say, we should have $b = 2ck$, that is, b even, contrary to hypothesis. Hence q is odd.

□

3. If a be prime to b , then—

- a) $(a + b)^m$ and $(a - b)^m$ have at most the G.C.M. 2^m ;
- b) $a^m + b^m$ and $a^m - b^m$ have at most the G.C.M. 2 ;
- c) $a + b$ and $a^2 + b^2 - ab$ have at most the G.C.M. 3 .

Lemma A. If x be prime to y , and g divide $x + y$, then g is prime to x and to y .

Proof. Let e be a common measure of g and x . Then e divides $x + y$ and x . Therefore e divides $(x + y) - x = y$ (§8). Thus e is a common measure of x and y . Because x and y are prime to each other, it follows e is unity. Hence g is prime to x , and in like manner g is prime to y . □

Lemma B. If x be prime to y , the G.C.M. of $x + y$ and $x - y$ divides 2.

Proof. Let g be the G.C.M. of $x + y$ and $x - y$. Then g divides $(x + y) + (x - y) = 2x$ (§8). Since g is prime to x (Lemma A), it follows g divides 2 (§10, Cor.1). $\square$

Proof of (a). Let g be the G.C.M. of $a + b$ and $a - b$. Then g divides 2 (Lemma B). Next, write $a + b = gp$ and $a - b = gq$ for integers p, q . Then p is prime to q ; for if they had a common factor e greater than unity, ge would be a common factor of $a + b$ and $a - b$ greater than their G.C.M. Hence p^m is prime to q^m (§10, Cor. 4), and integers A, B can be found such that $Ap^m - Bq^m = \pm 1$ (§14, Cor. 2). Thus $Ag^mp^m - Bg^mq^m = \pm g^m$. Now $(a + b)^m = g^mp^m$ and $(a - b)^m = g^mq^m$, so that every common factor of $(a + b)^m$ and $(a - b)^m$ divides this algebraic sum of their multiples (§8), that is, divides g^m . But g divides 2, and therefore g^m divides 2^m ; hence the G.C.M. of $(a + b)^m$ and $(a - b)^m$ is at most 2^m . $\square$

Proof of (b). Since a is prime to b , a^m is prime to b^m (§10, Cor. 4). Hence, by Lemma B applied to a^m and b^m , the G.C.M. of $a^m + b^m$ and $a^m - b^m$ divides 2, that is, is at most 2. $\square$

Proof of (c). Let g be the G.C.M. of $a + b$ and $a^2 + b^2 - ab$. Since g divides $a + b$ it divides $(a + b)^2$, and therefore divides $(a + b)^2 - (a^2 + b^2 - ab) = 3ab$ (§8). Since g divides $a + b$, it follows g is prime to a and b (Lemma A). Hence g is prime to their product ab (§10, Cor. 2). But g divides $3ab$ and is prime to ab , so g divides 3 (§10, Cor. 1), that is, the G.C.M. of $a + b$ and $a^2 + b^2 - ab$ is at most 3. $\square$

4. The difference of the squares of any two odd numbers is exactly divisible by 8.

Proof. Let a and b be odd, so that $a = 2k + 1$ and $b = 2l + 1$ for some integers k, l . Then

$$a^2 - b^2 = (4k^2 + 4k + 1) - (4l^2 + 4l + 1) = 4[(k^2 + k) - (l^2 + l)].$$

Now, k is either even or odd; if k is odd then k^2 is odd (Exercise 2, part 3) thus $k^2 + k$ is even (Exercise 2, part 1); if k is even, then k^2 and k are both divisible by 2, and so is their sum (§8), hence $k^2 + k$ is even. So we have in either case $k^2 + k$ is even. In likely manner $l^2 + l$ is even. That is, $k^2 + k = 2m$ and $l^2 + l = 2n$ for some integers m, n ; whence

$$a^2 - b^2 = 4(2m - 2n) = 8(m - n).$$

Therefore $a^2 - b^2$ is exactly divisible by 8. $\square$

5. The sum of the squares of three consecutive odd numbers increased by 1 is a multiple of 12.

Proof. Let a, b, c be three consecutive odd numbers, so that $a = 2k - 1$, $b = 2k + 1$ and $c = 2k + 3$ for some integer k . Then

$$\begin{aligned} a^2 + b^2 + c^2 + 1 &= (2k - 1)^2 + (2k + 1)^2 + (2k + 3)^2 + 1 \\ &= (4k^2 - 4k + 1) + (4k^2 + 4k + 1) + (4k^2 + 12k + 9) + 1 \\ &= 12k^2 + 12k + 12 \\ &= 12(k^2 + k + 1) \end{aligned}$$

Therefore $a^2 + b^2 + c^2 + 1$ is a multiple of 12. □

6. If each of two fractions be at its lowest terms, neither their sum nor their difference can be an integer unless the denominators be equal.

Proof. Let a/b and c/d be the fractions both at its lowest terms; that is, a is prime to b and c is prime to d . Suppose their sum to be an integer n so that

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm cb}{bd} = n$$

Hence $ad \pm cb = nbd$. Then $ad = nbd \mp cb = b(nd \mp c)$, so b divides ad , since a and b are prime to each other, we have that b divides d . Similarly $cb = nbd \mp ad = d(nb \mp a)$ so d divides cb , since d is prime to c , we have that d divides b . Because d divides b and b divides d they must be equal, that is $b = d$. □

7. Resolve 45738 and 297675 into their prime factors.

$$\begin{aligned} 45738 &= 2 \times 3^3 \times 7 \times 11^2 \\ 297675 &= 3^5 \times 5^2 \times 7^2 \end{aligned}$$

8. Find the G.C.M. of 54643 and 91319, using negative remainders whenever it is of advantage to do so.

$$\begin{aligned} 91319 &= 54643 \times 2 - 17967 \\ 54643 &= 17967 \times 3 + 742 \\ 17967 &= 742 \times 24 + 159 \\ 742 &= 159 \times 5 - 53 \\ 159 &= 53 \times 3 \end{aligned}$$

The G.C.M is 53.

9. Prove that the L.C.M. of two integers is the quotient of their product by the G.C.M.

Proof. Let g be the G.C.M. of a and b , and write $a = g\alpha$, $b = g\beta$. Now $ab/g = g\alpha\beta$, and $g\alpha\beta = (g\alpha)\beta = a\beta$, $g\alpha\beta = (g\beta)\alpha = b\alpha$, so that $g\alpha\beta$ is a common multiple of a and b . Let M be any common multiple of a and b , say $M = ak = g\alpha k$. Since b divides M , $g\beta$ divides $g\alpha k$, that is, β divides αk ; and β , being prime to α , divides k (§10, Cor. 1). Write $k = \beta t$; then $M = g\alpha\beta t$. Hence every common multiple of a and b is a multiple of $g\alpha\beta$, which is therefore the least of them, that is, the L.C.M. of a and b is ab/g . $\square$

10. If g_1, g_2, g_3 be the G.C.M.'s, l_1, l_2, l_3 the L.C.M.'s, of b and c , c and a , a and b respectively, G the G.C.M., and L the L.C.M., of the three a, b, c , show that

$$\text{a) } L = \frac{abcG}{g_1g_2g_3};$$

$$\text{b) } \frac{L}{G} = \sqrt{\frac{l_1l_2l_3}{g_1g_2g_3}}.$$

11. When x is divided by y , the quotient is u and the remainder v ; show that, when x and uy are divided by v , the remainders are the same, and the quotients differ by unity.